PROJECT-BASED LEARNING

PHASE-I EVALUATION

TOPIC: ***Multi-Drone Search & Rescue for Dynamic Disaster Response Using Graph-Based Path Planning and Priority Task Allocation***

Team ID: DSCPP-III-2026-T290

Department of Computer Science & Engineering
Graphic Era (Deemed to be University), Dehradun

Academic Session 2026–27

Team & Mentor Details

Team Information

TeamID: DSCPP-III-2026-T290
Semester: 3rd
Section: DS2, DS1, J, CYBER
Domain: Artificial Intelligence & Unmanned Aerial Vehicle Systems- DS(C) AND OOPS(C++)

Team Members

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Mentor

(Prof).Dr.Amit Kumar
Interactions completed: _____02_____

Problem Statement & Motivation

Problem Statement

Search-and-rescue teams often face large, hazardous areas where manually surveying every location is slow and difficult.
The proposed system coordinates multiple UAVs by assigning high-priority search tasks to suitable drones while considering battery level, distance, capability and availability.
It also plans efficient routes and reassigns tasks when conditions change.

Motivation

Current SARoperations may involve limited information, changing priorities and resource constraints. A coordinated software layer can reduce duplicated searches, improve task distribution and support faster decision-making.

Target Users / Stakeholders

Primary users: disaster-response coordinators and SAR teams. Secondary stakeholders: emergency agencies and UAV operators. Expected benefit: better coverage, resource utilization and response time.

Objectives

01 Model a disaster search area as a graph containing SAR locations, search zones, obstacles and possible routes.

02 Prioritize rescue missions using a priority-based task queue and allocate suitable UAVs using measurable constraints.

03 Use shortest-path algorithms to generate efficient routes and avoid unavailable or blocked paths.

04 Implement the system with C for data-structure logic and C++ OOP for UAV, mission, zone and coordination modules.

05 Evaluate performance using test scenarios such as low battery, UAV failure, new high-priority missions and route blockage.

Key Objective: Develop a working multi-UAV coordination simulation that improves mission assignment and route efficiency.

Proposed Solution

1. Input
UAV status, search zones, priorities, map and constraints

→

2. Pre-process
Validate data and build graph / mission records

→

3. Prioritize
Use priority queue to rank rescue tasks

→

4. Allocate
Score UAVs by priority, distance, battery and capability

→

5. Plan Route
Use BFS / Dijkstra to generate an efficient path

→

6. Monitor
Update battery, failures, new tasks and blocked routes

→

7. Output
Show assignments, routes, distances and status

Key Features

- Priority-based mission queue with dynamic reassignment.
- UAV suitability scoring using battery, distance and capability.
- Graph-based shortest-path planning with blocked-route handling.
- Failure / low-battery simulation and automatic task reassignment.
- Optional validation through ArduPilot SITL after the core C/C++ simulator is stable.

Practical implementation: a controlled C/C++ simulation; later connect the planning logic to a software-in-the-loop UAV simulator if required.

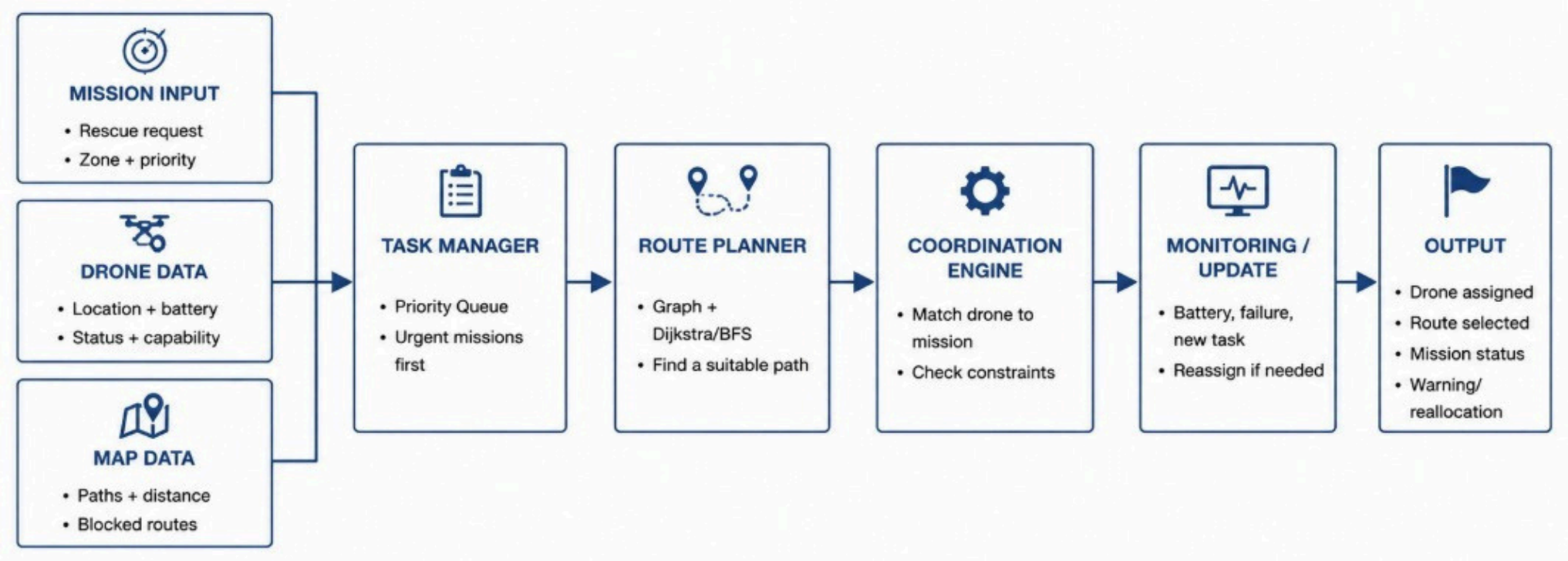
Integration of PBL Course Concepts

Course	Concepts Applied	Project Module / Use
Data Structures in C	Graphs, adjacency lists, queues, priority queues, arrays, searching and sorting. These are used to represent the disaster map, manage rescue missions, find suitable routes and rank UAVs based on priority and availability.	Used in Map Representation, Route Planning, Mission Queue and UAV Ranking. Graphs help find routes between locations, priority queues handle urgent missions first, and searching/sorting help select suitable UAVs.
OOPs with C++	Classes, objects, encapsulation, inheritance, polymorphism and STL containers. These concepts are used to create and manage different components of the drone coordination system.	Used in UAV, Mission, SearchZone, Route, Coordinator and Simulation modules. OOP helps manage UAV properties, missions and routes while coordinating multiple drones efficiently.

Why the integration is meaningful

Chandles the core datastructures andalgorithmic operations, while C++ OOP organizes the system into reusable real-world entities.
Together they form oneworkflow: represent prioritize allocate route simulate evaluate.

System Architecture / Workflow



Control Flow

1)Receive mission and environment data
2) update graph and UAV states
3) rank rescue tasks
4) calculate UAV-task suitability
5) assign missions
6) calculate routes
7) simulate execution and update status
8) reallocate when a constraint changes.

Practical deployment path: C/C++ simulation visualization optional ArduPilot SITL validation.

Technology Stack & Methodology

Technology Stack

- C— data structures andalgorithms
- C++ — OOPsystemmodules
- STL — vectors, priority_queue, maps
- IDE — VS Code / Code::Blocks
- Data — in-memory structures + CSV/text test files
- Optional simulation — ArduPilot SITL / Mission Planner

Development Methodology

1. Requirement analysis
2. System and graph design
3. C/C++ module implementation
4. Test scenarios & fault cases
5. Performance evaluation
6. Simulation validation & documentation

Technology justification

C/C++directly matches thePBL subjects and is sufficient for the core simulation. A simulator such as ArduPilot SITL is optional for later validation because it can run virtual vehicles without physical hardware and can emulate failures and sensor conditions.

Initial Progress & Team Contribution

Progress Achieved

- Topic selected and feasibility discussed with mentor.
- Basic background study completed on UAV search-and-rescue and path planning.
- Project scope finalized as a software-based multi-drone coordination system.
- Initial system flow and module structure prepared.
- DSA approach selected: graph, priority queue, shortest-path routing and sorting.
- OOP entities identified: Drone, Mission, SearchZone, Route and Coordinator.

Member	Role	Current Contribution
Hrimandeep Singh	Project / Docs	Requirements, presentation structure and documentation.
Jayna Kothari	DSA / Algorithms	Graph model, priority queue and route-planning logic.
Aarav Fangalia	OOP / Design	C++ classes, modules and simulation entities.
Raghav Singhal	Research / Testing	Background study, feasibility and test scenarios.

Progress reflects Phase-I work; implementation and testing will continue in later phases.

Roadmap, Expected Outcomes & References

Roadmap

Phase-I Topic selection, feasibility study, basic research and system design.

Phase-II C/C++ implementation of data structures, task allocation and route planning.

Phase-III Testing with multiple missions, battery limits, failures and changing priorities.

Expected Outcomes

• WorkingC/C++ prototypefor drone task allocation.
• Graph-based route planning and shortest-path output.
• Priority-based mission scheduling.
• Handling of low battery, unavailable drones and new missions.
• Simple simulation results for different rescue scenarios.

References & Feasibility Sources

1.ArduPilotDocumentation—Simulation/SITL. Usefulfor understanding software-baseddrone simulationwithout physical hardware.

2. GeeksforGeeks — Dijkstra's Shortest Path Algorithm. Basic reference for graph- based route planning.

3. cppreference — C++ Classes, Encapsulation, Inheritance and Polymorphism. Reference for OOP implementation.

- Task allocation and path planning for multi-UAV search and rescue in complex environments — Simulation Modelling Practice and Theory, 2026
- Multiple UAV Coordination for Enhanced Search and Rescue Missions — IFAC-PapersOnLine, 2025

Feasibility:
The project starts as a C/C++ software simulation; SITL can be considered later if needed.